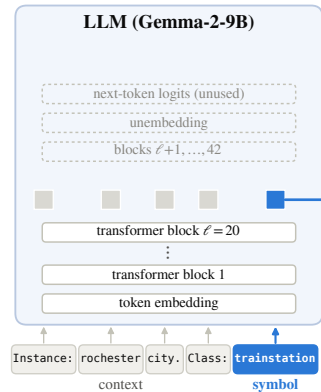
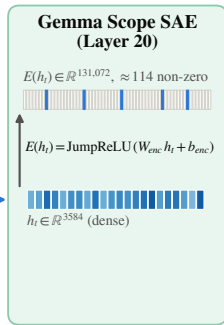


1 Symbol in context



2 SAE encoder



3 Sparse code, rarity-weighted

$a(s) = \text{mean of } E(h_t) \text{ over symbol tokens and contexts}$

$a_f(s)$	$w_f(s) = a_f(s) \cdot \text{idf}_f$
40.38	23.95
"open source software license..."	df 146/266
33.50	0.00
"sections of text related to..."	df 266/266
16.55	65.75
"references to railroads and..."	df 4/266
13.37	0.00
"technical terms and syntax..."	df 266/266
13.05	63.86
"words related to tracking or..."	df 1/266
12.27	42.99
"references to 'stations' and..."	df 7/266

...: feature meaning (automatic explanation)

4 Matching

Query (NELL): **trainstation**
Gold (DBpedia): **RailwayStation**

SAE-IDF top 1: RailwayStation
(cos sim.(query, top 1) = 0.37, rank = 1)

60% "references to railroads and..."

32% "references to 'stations' and..."

3% "terms related to transportation hubs and..."

1% "specific terminologies related to..."

Dense top 1: AutomobileEngine

(SAE-IDF: cos sim.(query, top 1) = 0.08, rank = 4)

25% "open source software license text and..."

16% "words with scientific or technical suffixes..."

9% "medical and clinical terminology related to..."

8% "terms related to ecosystems and..."